

Internet Service Churn Analysis



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December 7 2025

RevoU FSDA Batch OCT25

This deck is a churn analysis in the internet services industry, which focuses on understanding the reasons why customers switch services.

Churn is driven by usage friction & experience—not price.

1. Project Description

Churn analysis for MyIndiNet (B2C ISP) to identify the main drivers of customer churn and prioritize the highest-impact retention levers.

2. Objectives

Reduce churn from 55.4% → <50% within 6 months (Q1 2026) by addressing the top 3 churn drivers.

3. Tools

- ❖ Google Colab (Python) for cleaning + EDA + modeling
- ❖ Tableau for interactive dashboards
- ❖ Keynotes for storytelling & pitching

4. Metrics

- ❖ Churn Rate, Retention Rate
- ❖ Download Over-Limit Frequency

- ❖ Service Failure Count
- ❖ Bundle Adoption (TV/Movie)
- ❖ (Just a small option: ROC-AUC model)

5. Analysis Results

- ❖ Over-limit usage is the strongest churn trigger: churn reaches 99.8% at high over-limit frequency.
- ❖ Bundling is churn protection: TV bundle customers churn ~42% lower than non-bundle customers.
- ❖ Longer remaining contracts reduce churn risk; pricing (bill avg) is not the main driver.
- ❖ (This is all in your deck: churn 55.4%, overlimit 99.8%, bundling 42% diff, and insight experience>price.)

6. Recommendations

- ❖ Redesign over-limit policy: proactive alerts + flexible add-ons / unlimited plans.
- ❖ Targeted bundling upsell: TV/Movie offers for high-risk non-bundle users.
- ❖ Early renewal program: incentives 2–3 months before contract ends + loyalty perks.

MyIndiNet is a B2C internet provider where retention directly impacts recurring revenue customizable internet packages to individual customers.



Business Models :

- *Monthly subscription revenue*
- *Additional services (TV bundle, movie package, premium support)*

Key Business Focus :

- *Customer acquisition*
- *Service quality*
- *Churn management & retention*

Why Churn Matters?

High churn directly reduces recurring revenue and increases customer acquisition costs.



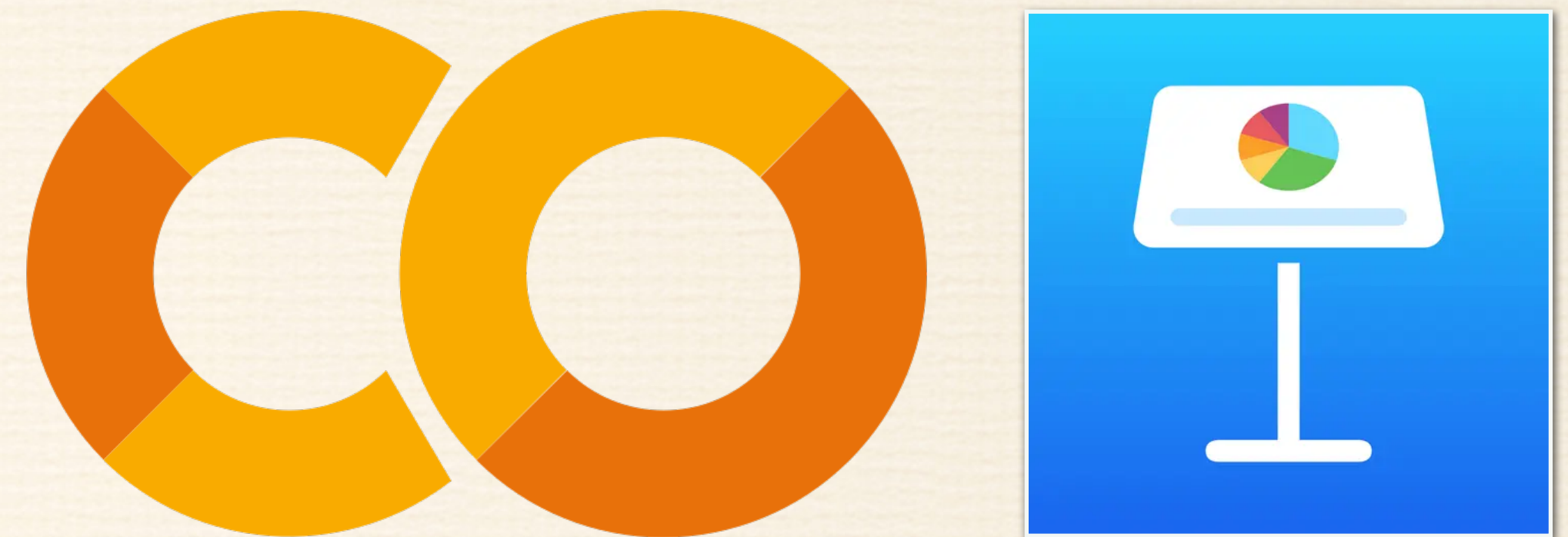
A structured data-driven approach was used to identify churn drivers

Data Collection

- Data Cleaning
- Exploratory Analysis
- Predictive Modeling
- Insights & Recommendations

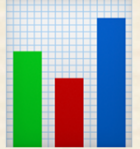
Tools:

- ❖ Python (Google Colab)
- ❖ Tableau
- ❖ Keynote



+tableau++public

Business Questions

 What are the main factors driving customer churn?

 How does usage behavior impact churn rates?

 How do service issues affect customer retention?

 Which customer segments are most at risk?

MyIndiNet must urgently address churn to protect recurring revenue

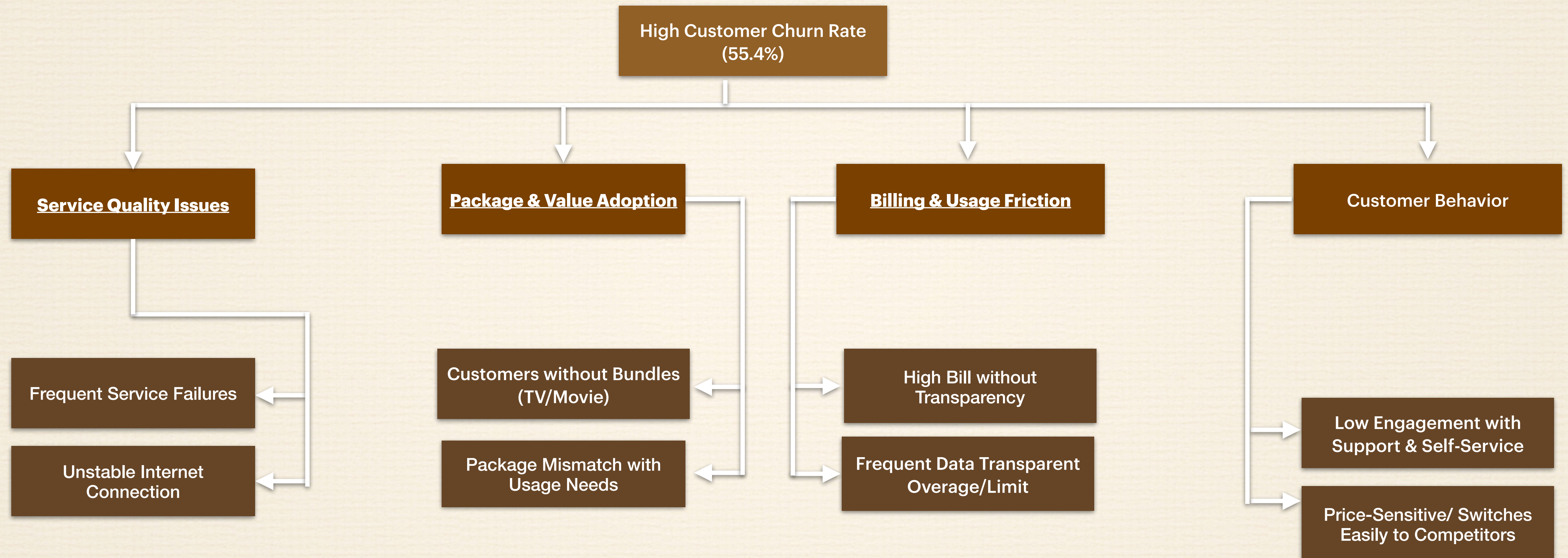
Problem Statement:

How can MyIndiNet reduce its churn rate from 55.4% to below 50% within 6 months (Q1 2026) by identifying and addressing the key drivers related to customer experience, usage friction, and subscription features?

Objective :

- ❖ Identify major churn drivers
- ❖ Develop data-driven retention strategies
- ❖ Improve customer retention within 6 months

Churn is driven by service experience, usage friction, and weak value adoption



Hypothesis

H1 | Service Quality

If customers experience high service failure counts, their probability of churn will be higher than customers with minimal disruptions.

H2 | Bundling Adoption

If customers subscribe to a bundled package (TV/Movie) that suits their needs, their churn rate will be lower than customers without a bundle.

H3 | Billing & Usage Friction

If customers have high monthly bills and frequently exceed their download quota, they are more likely to churn due to an imbalanced perception of value.

Data Overview

Dataset: churn_cleaned_final.csv

- ❖ Size: 72,274 records × 11 columns
- ❖ Target Variable: Churn (binary: 0=retained, 1=churned)

Key Features:

- ❖ Subscription: TV subscriber, Movie package
- ❖ Usage: Download avg, Upload avg, Download over limit
- ❖ Contract: Remaining contract length
- ❖ Service: Service failure count, Bill avg
- ❖ Engagement: Subscription age
- ❖ Data Quality: No missing values, no duplicates, cleaned

Data Schema (Data Dictionary)

Column	Data Type	Description	Role in Churn Analysis
id	Integer	Unique customer ID	Primary key
istvsubscriber	Binary (0/1)	TV package subscriber?	Bundling insight (-42% churn)
ismoviepackagesubscriber	Binary (0/1)	Movie package subscriber?	Bundling insight
subscriptionage	Float	Months since subscription start	Engagement proxy
billavg	Float	Average monthly bill	Price sensitivity check
reaminingcontract	Float	Months left on contract	Contract length insight
servicefailurecount	Float	Number of service failures	Service quality insight
downloadavg	Float	Average monthly download (GB)	Usage pattern
uploadavg	Float	Average monthly upload (GB)	Usage pattern
downloadoverlimit	Integer	Times exceeded download limit	#1 Churn driver (99.8% at 7x)
churn	Binary (0/1)	Target: Churned (1) or Retained (0)	Objective variable

Data Cleaning & Preparation

Data Quality Assessment

Foundation Datasets

- ❖ 72,274 customer records, 11 features analyzed
- ❖ All data types verified: int64 (ID, bill_avg, service_failure_count), uint8 (binary flags), float64 (continuous metrics)

Data Integrity

- ❖  Missing Values: 0 across all columns — no imputation required, dataset complete
- ❖  Duplicate Records: 0 duplicates found — all customer IDs unique

No data removal or handling procedures required

Outlier Assessment

Identified in high-usage features via percentile analysis (1%, 5%, 95%, 99%):

- ❖ bill_avg: Max 406 (99th percentile: 59) — high-bill segment
- ❖ download_avg: Max 4415 (95th percentile: 139.2) — power users
- ❖ upload_avg: Max 453 (95th percentile: 13.1) — heavy uploaders

Decision: Retained for modeling (outliers represent legitimate high-usage customers, relevant to churn prediction)

	missing_count	missing_pct
id	0	0.0
is_tv_subscriber	0	0.0
is_movie_package_subscriber	0	0.0
subscription_age	0	0.0
bill_avg	0	0.0
reamining_contract	0	0.0
service_failure_count	0	0.0
download_avg	0	0.0
upload_avg	0	0.0
download_over_limit	0	0.0
churn	0	0.0

	0
id	int64
is_tv_subscriber	uint8
is_movie_package_subscriber	uint8
subscription_age	float64
bill_avg	int64
reamining_contract	float64
service_failure_count	int64
download_avg	float64
upload_avg	float64
download_over_limit	uint8
churn	uint8

	count	mean	std	min	1%	5%	50%	95%	99%	max
subscription_age	72274.0	17.000000	0.000000	17.0	17.0	17.0	17.0	17.0	17.000	17.0
bill_avg	72274.0	18.942483	13.215386	0.0	0.0	0.0	19.0	34.0	59.000	406.0
service_failure_count	72274.0	0.274234	0.816621	0.0	0.0	0.0	0.0	2.0	4.000	19.0
download_avg	72274.0	43.459595	63.317706	0.0	0.0	0.0	27.5	139.2	236.927	4415.2
upload_avg	72274.0	4.169977	9.797685	0.0	0.0	0.0	2.1	13.1	35.000	453.3
download_over_limit	72274.0	0.207613	0.997123	0.0	0.0	0.0	0.0	1.0	6.000	7.0

Exploratory Data Analysis — Churn Overview

📊 Overall Churn Rate: 55.4% (40,050 from 72,274 customers)

💠 Churn by Key Features:

❖ TV Subscribers: No=89.6%, Yes=47.7%

❖ Movie Package: No=66.2%, Yes=33.9%

❖ Download Over Limit: 0x=53.5%, 7x=99.8% ⚠️

Metric	Churned	Active
Subscription Age	2.2	2.7 thn
Bill Avg	\$18	\$19
Service Failures	0.29	0.25

	subscription_age			bill_avg			service_failure_count		
	mean	median	std	mean	median	std	mean	median	std
churn									
0	2.727828	2.05	2.348857	18.765051	18.0	8.323764	0.254593	0.0	0.751527
1	2.226554	1.97	1.709438	18.358502	20.0	10.902273	0.290037	0.0	0.865134

```
(churn
0    32224
1    40050
Name: churn, dtype: int64,
np.float64(0.5541411849351081))
```

```
{'is_tv_subscriber': is_tv_subscriber
0    0.896195
1    0.476630
Name: churn, dtype: float64,
'is_movie_package_subscriber': is_movie_package_subscriber
0    0.662418
1    0.338846
Name: churn, dtype: float64,
'download_over_limit': download_over_limit
0    0.535284
1    0.736292
2    0.812500
3    0.895582
4    0.929825
5    0.925408
6    0.962209
7    0.998016
Name: churn, dtype: float64}
```


Feature Analysis — Usage & Engagement Patterns

◆ Usage Distribution (Download & Upload):

- ❖ Download: Mean 42 Mbps, Max 237 Mbps (capped p99)
- ❖ Upload: Mean 3.8 Mbps, Max 35 Mbps (capped p99)

◆ Service Quality Impact:

- ❖ Avg service failures: 0.27 per customer
- ❖ Correlation: More failures → Higher churn risk

◆ Subscription Tenure:

- ❖ Avg tenure: 2.45 years
- ❖ Churned customers: Slightly shorter tenure (2.2 vs 2.7 years)

	subscription_age			bill_avg			service_failure_count		
	mean	median	std	mean	median	std	mean	median	std
churn									
0	2.727828	2.05	2.348857	18.765051	18.0	8.323764	0.254593	0.0	0.751527
1	2.226554	1.97	1.709438	18.358502	20.0	10.902273	0.290037	0.0	0.865134

Churned customers had shorter subscription periods (2.23 vs. 2.73 years) and slightly more service interruptions. Average bills were nearly identical, meaning churn was driven more by experience/service than bill size.

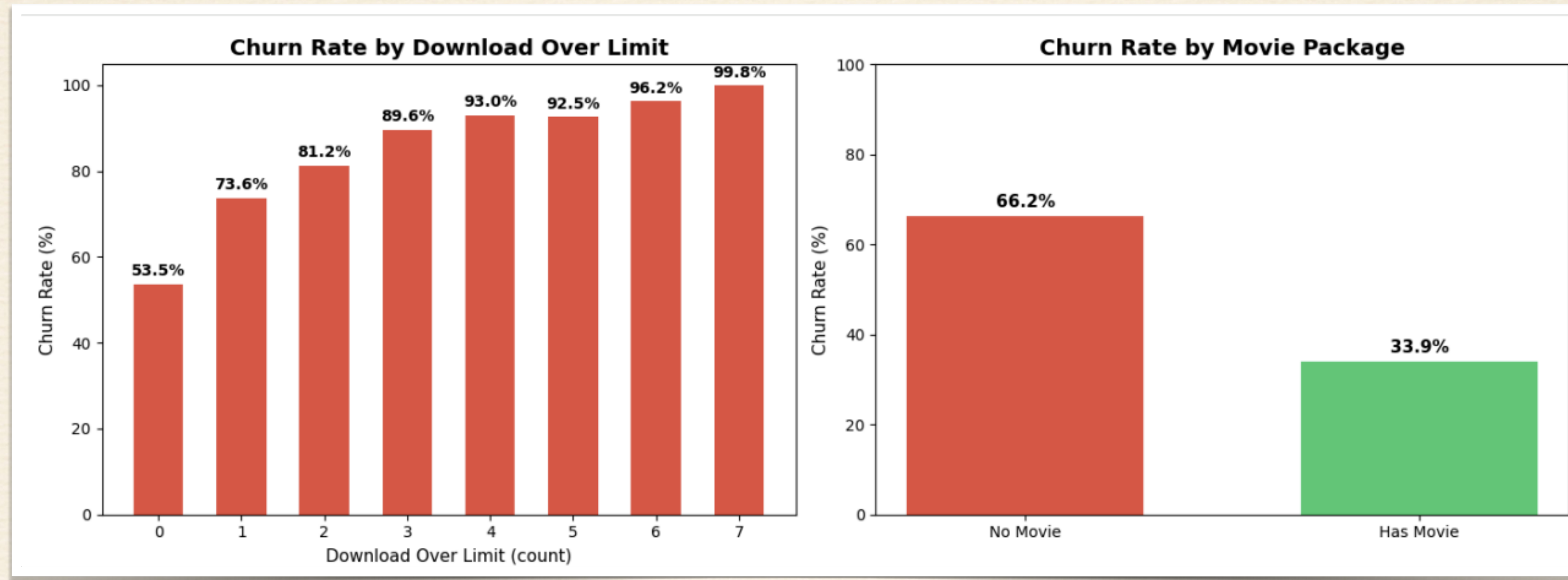
Model Insights — Predictive Risk Factors

	0
download_over_limit	0.590672
service_failure_count	0.138480
bill_avg	0.021407
upload_avg	-0.006194
download_avg	-0.017758
subscription_age	-0.168958
is_movie_package_subscriber	-0.288720
is_tv_subscriber	-1.863328
reamining_contract	-2.656895

"The strongest features driving churn are download_over_limit and service_failure_count, while remaining_contract, is_tv_subscriber, and is_movie_package_subscriber significantly reduce the risk of churn."

"This means that over-limit penalties and service disruptions are the main triggers for customer churn, while long contracts and TV/movie bundling act as 'churn protection'."

Churn by Binary Features



❖ TV subscribers: 89.6% (No TV) vs 47.7% (Has TV) — 42% difference!

❖ Movie package: 66.2% (No Movie) vs 33.9% (Has Movie) — bundling reduces churn significantly

❖ Download over-limit: Strong escalation — 53.5% (0x) → 99.8% (7x over-limit events)

❖ Key insight (bold): Bundling packages (TV/Movie) act as churn protection; frequent over-limit penalties are the strongest churn trigger

Customer Characteristics: Active vs Churned

Metric	Active Customers	Churned Customers
Subscription Age (years)	2.73	2.23
Bill Average (\$)	18.77	18.36
Service Failures (count)	0.25	0.29

- ❖ Churned customers have shorter tenure (2.23 vs 2.73 years) — less loyalty
- ❖ Bills are nearly identical (~\$18–19) — churn **NOT** driven by price alone
- ❖ Service failures slightly higher for churned customers (0.29 vs 0.25) — service quality matters
- ❖ Key insight (bold): Churn is driven more by experience/behavior (failures, over-limits, bundling) than by billing amount

Predictive Model: Logistic Regression Baseline

Setup Models:

- ❖ Type: Logistic Regression (binary classification)
- ❖ Features: 9 predictors (TV subscribers, Movie package, download_over_limit, subscription_age, bill_avg, remaining_contract, service_failure_count, download_avg, upload_avg)
- ❖ Data split: 80% train, 20% test (stratified by churn)

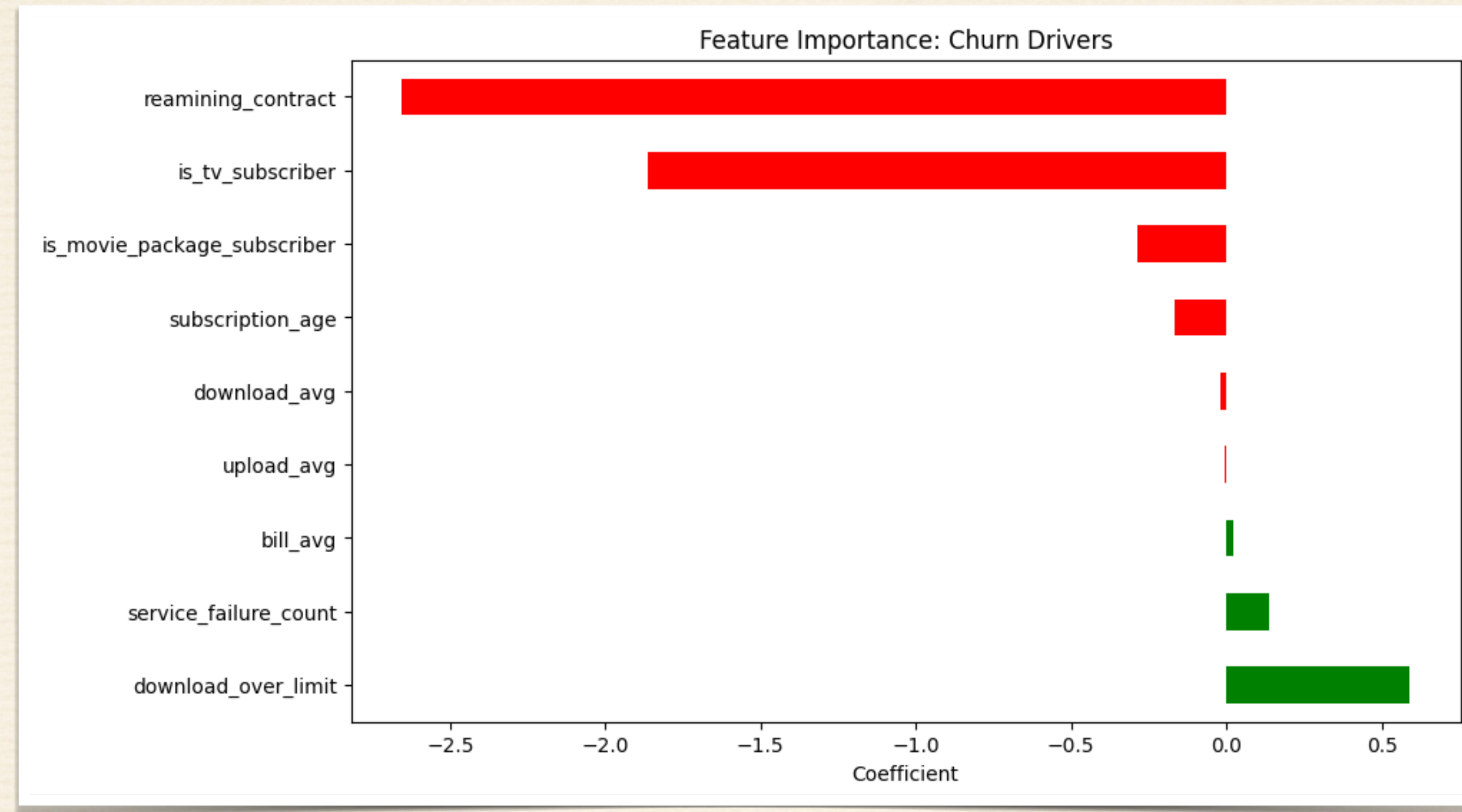
Performance Metrics:

- ❖ Accuracy: 84%
- ❖ Precision (Churn): 83%
- ❖ Recall (Churn): 88%
- ❖ ROC-AUC: 0.90 (excellent discrimination between churn/non-churn)

Key insight:

Model achieves strong predictive performance; can reliably identify high-risk churn customers for targeted retention campaigns

Feature Importance & Churn Drivers



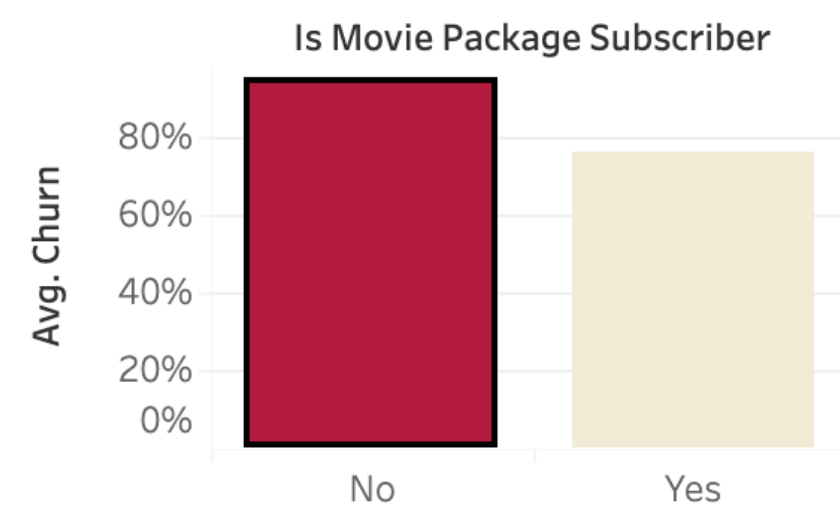
- ❖ Strongest churn trigger: download_over_limit (+0.59) & service_failure_count (+0.14).
- ❖ Strongest churn protection: remaining_contract (-2.66) & is_tv_subscriber (-1.86).
- ❖ Takeaway: Reduce over-limit friction + improve service stability; push bundling & early renewal.

Dashboard

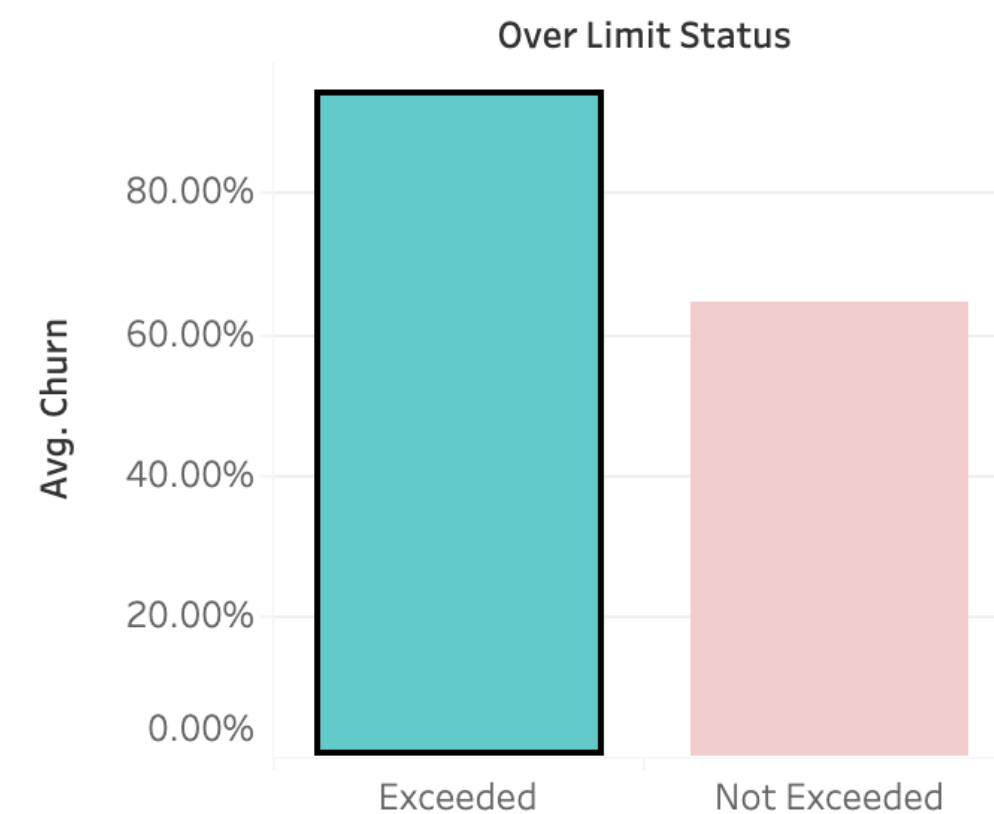
Overall
Churn Rate

93.7%

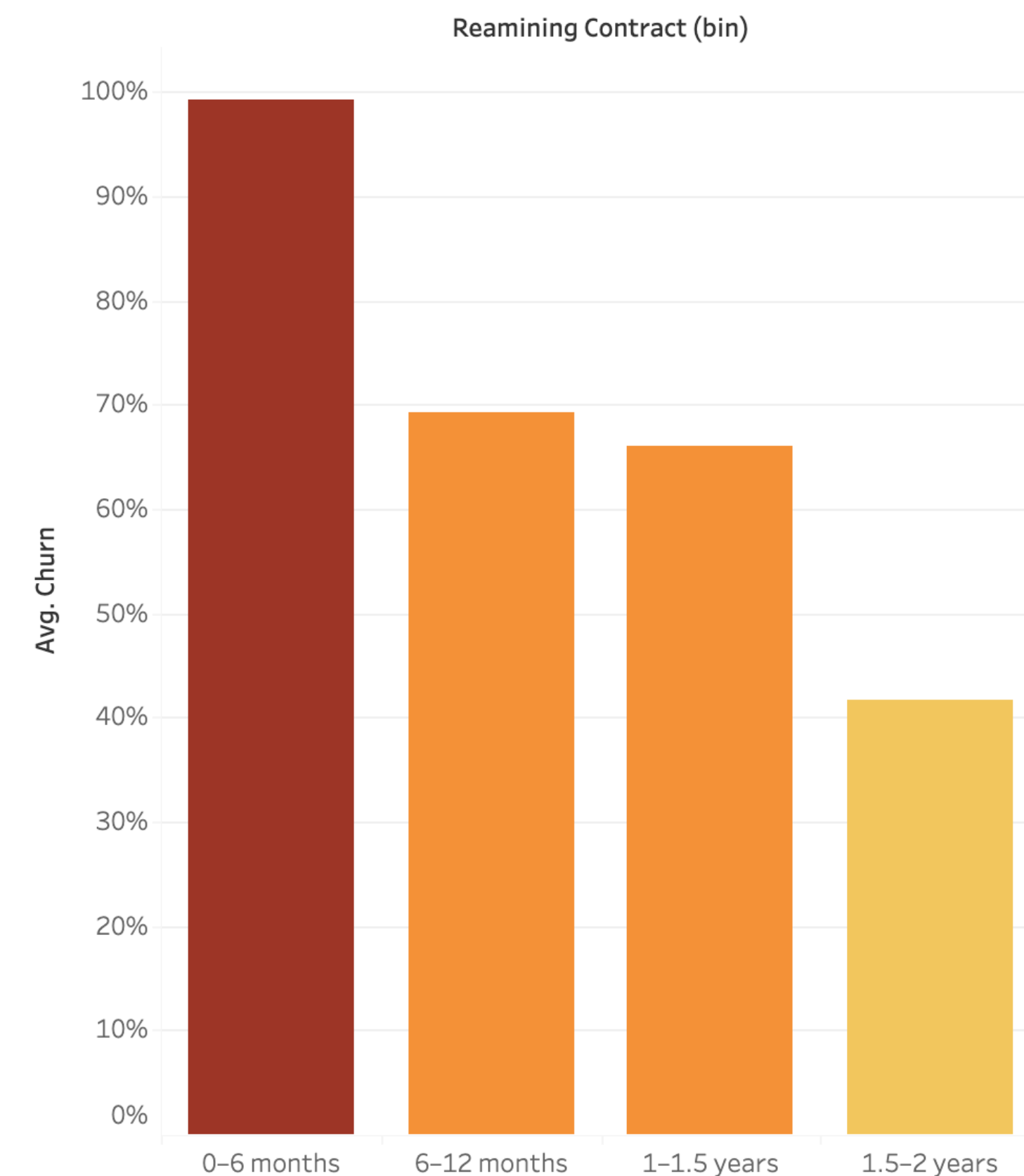
Churn Rate by Movie Package
Subscription



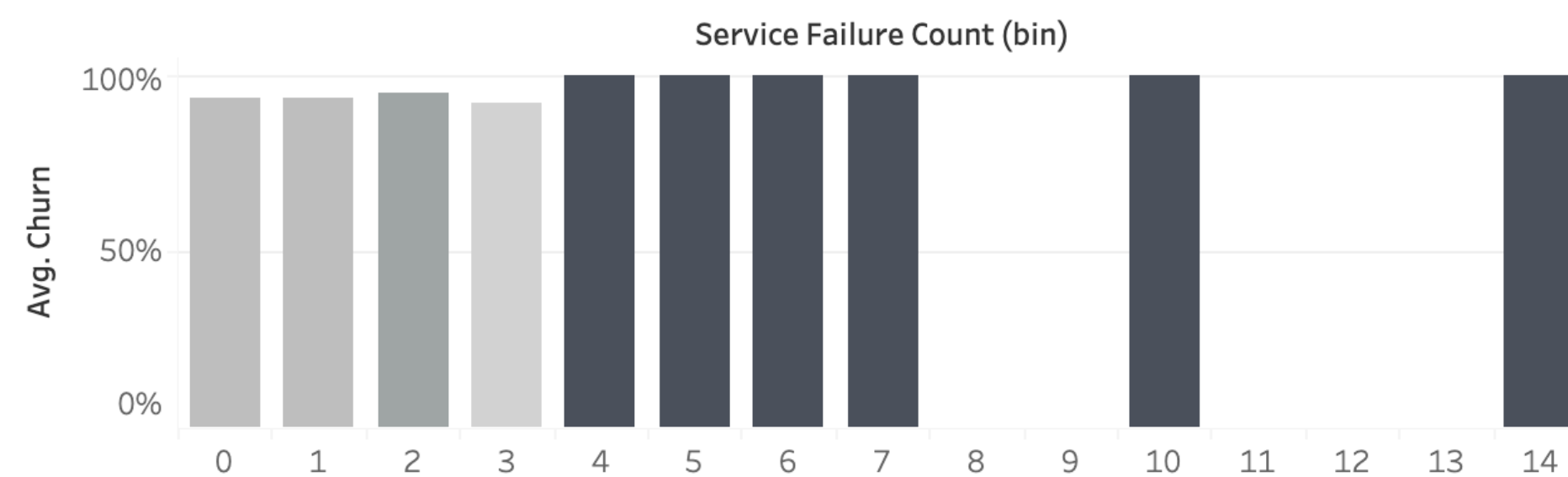
Churn Rate by Download Limit
Exceedance



Churn Rate by Remaining Contract Duration



Churn Rate by Service Failure Frequency



Current Conditions

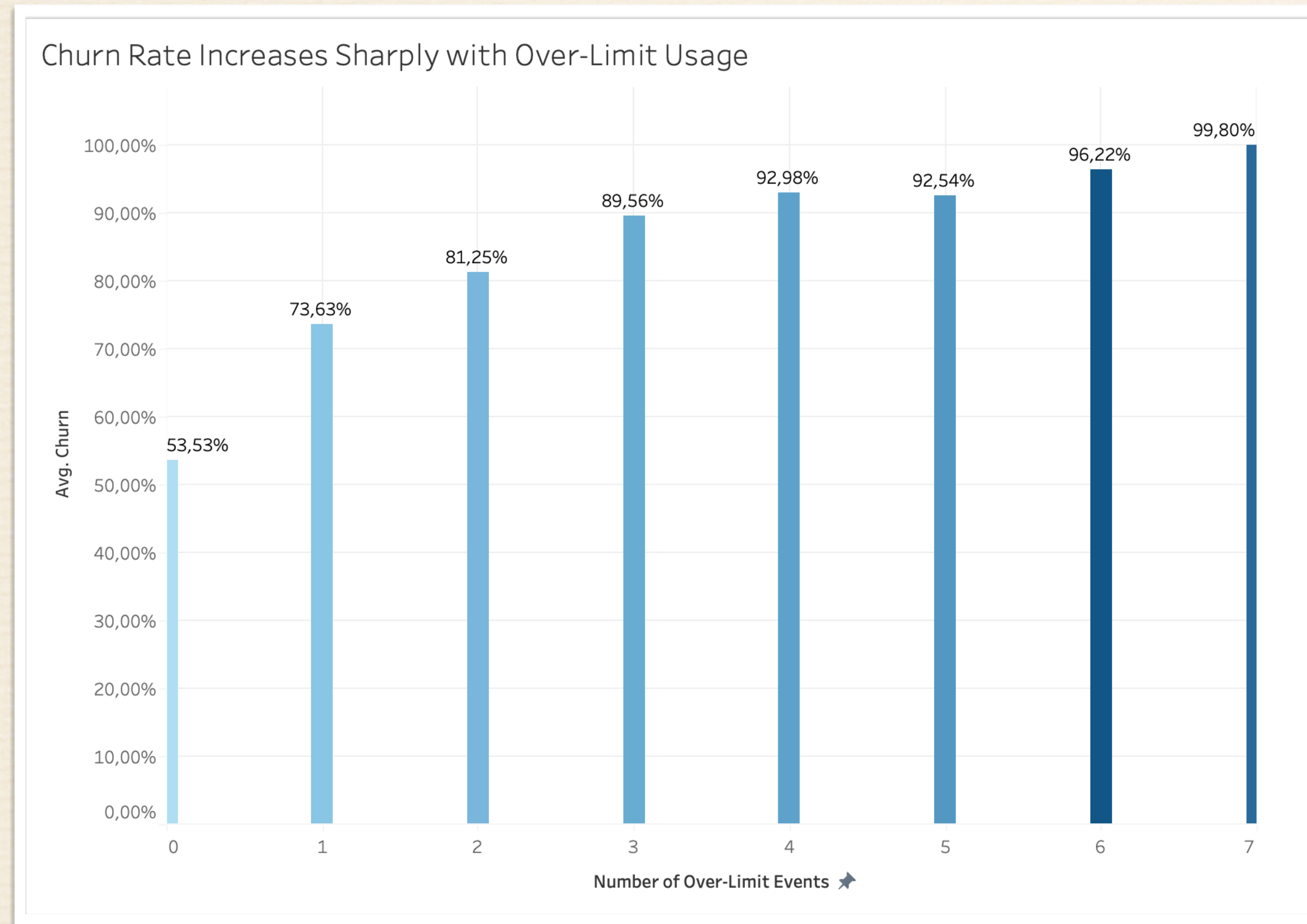
More than half of customers are leaving MyIndiNet.

Overall Churn Rate

55,4%

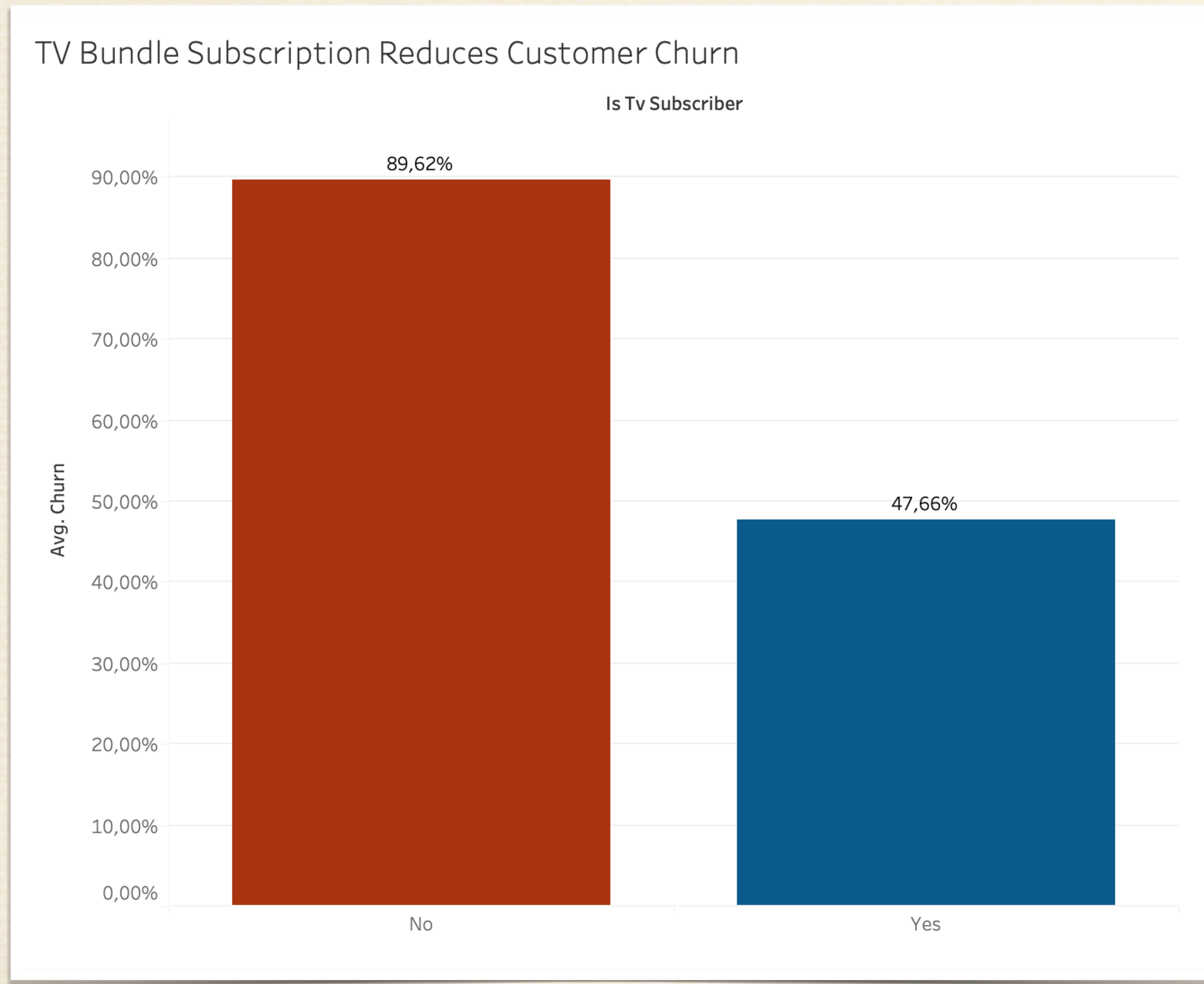
Over 40,000 out of 72,274 customers have churned, indicating a critical retention issue.”

Exceeding download limits triggers almost all customer churn



- ❖ Customers who frequently exceed download limits show nearly 100% churn rate, indicating severe usage friction and perceived unfair billing policies.

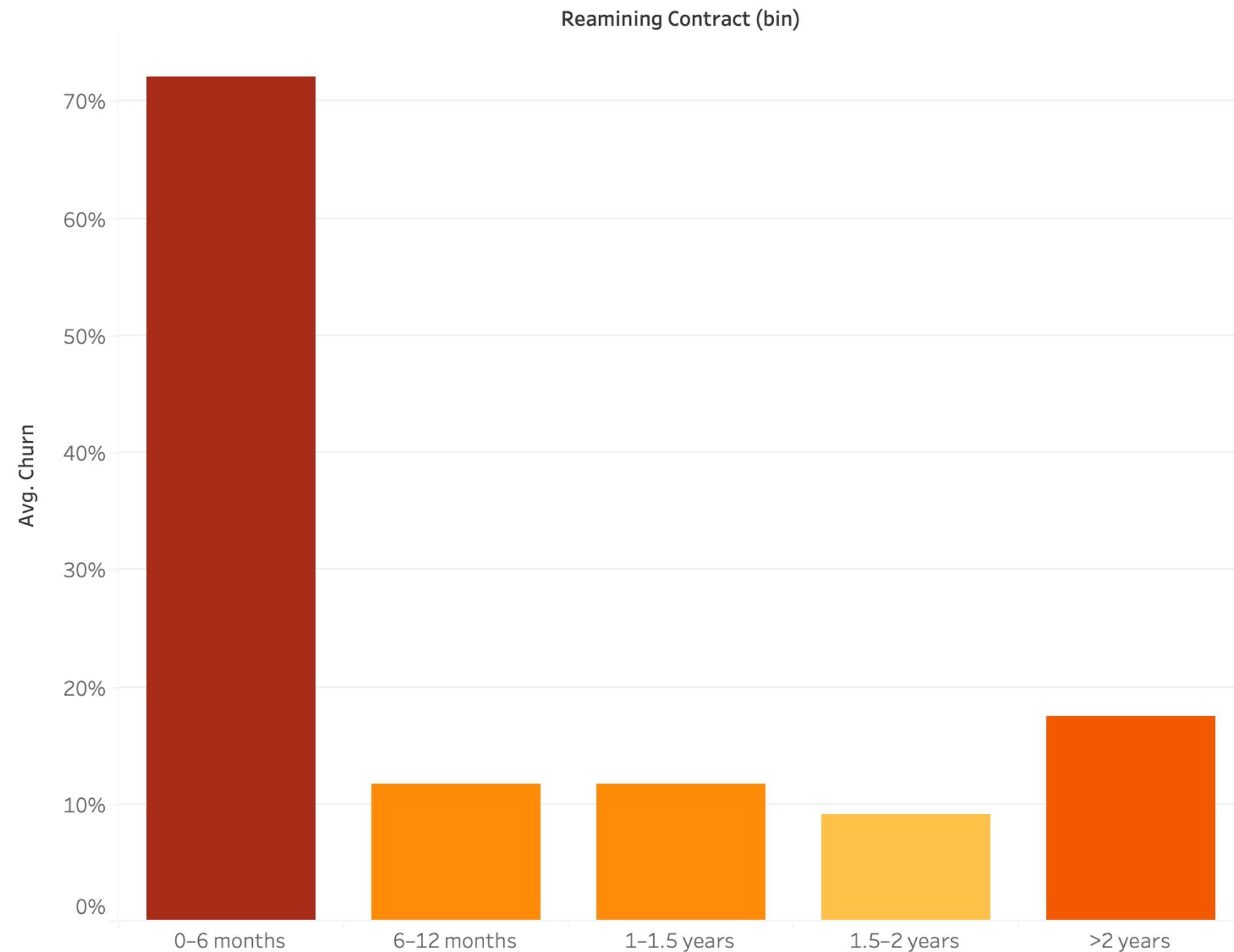
Customers with bundled services churn significantly less



- ❖ Customers without TV bundles experience extremely high churn (~89.6%)
- ❖ Bundled services cut churn almost in half (down to ~47.7%)

Remaining Contract Duration Strongly Impacts Churn

Churn Rate by Remaining Contract Duration

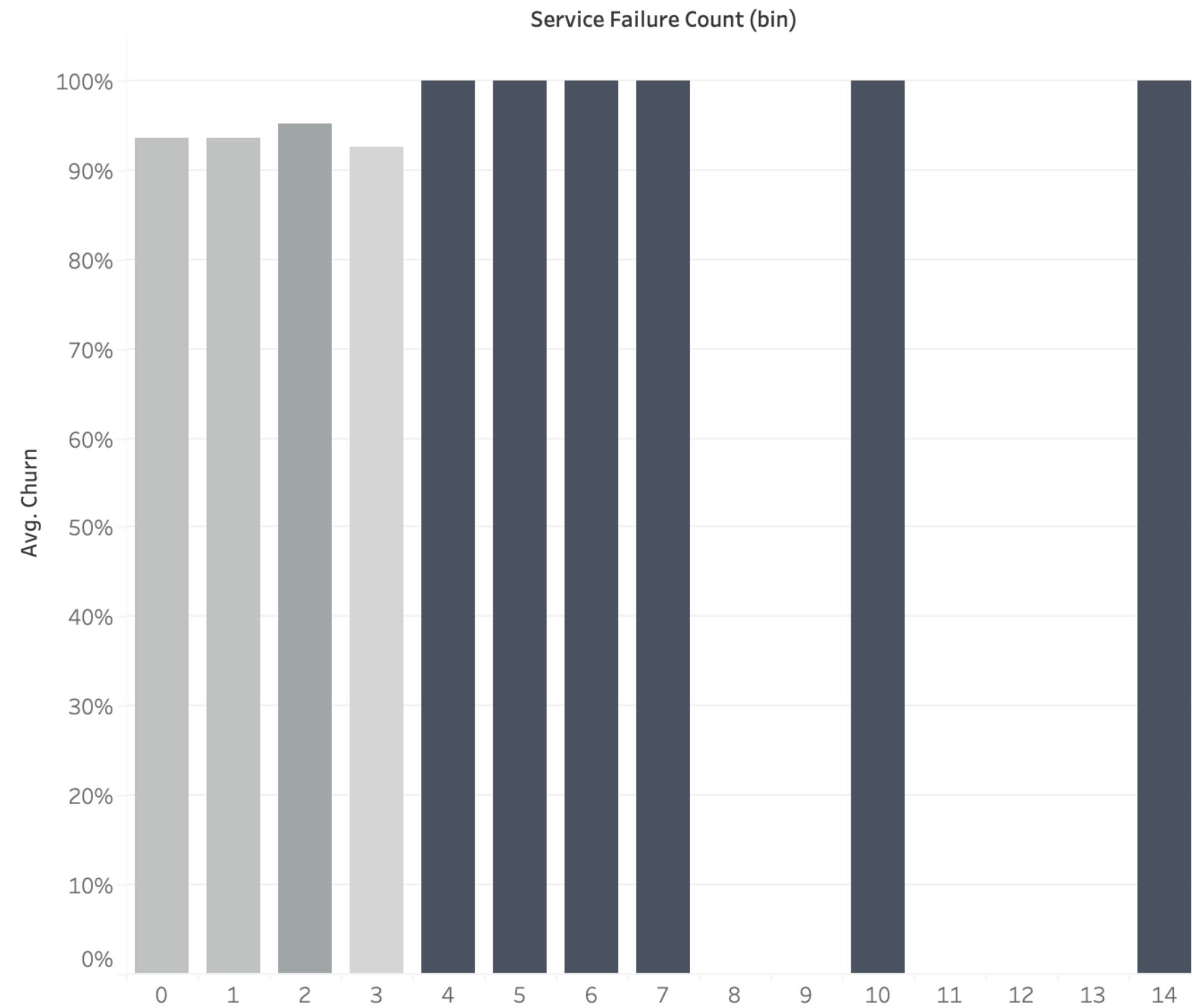


Customers with less than 6 months remaining on their contracts show the highest churn rate (above 70%), while longer contract durations significantly reduce churn risk.

- ❖ The highest churn occurs when the contract has 0–6 months remaining.
- ❖ After 6 months, churn drops dramatically to around 10–18%.
- ❖ Longer contracts → much more stable customers.

Service failures significantly increase customer churn

Churn Rate by Service Failure Frequency



Key insight:

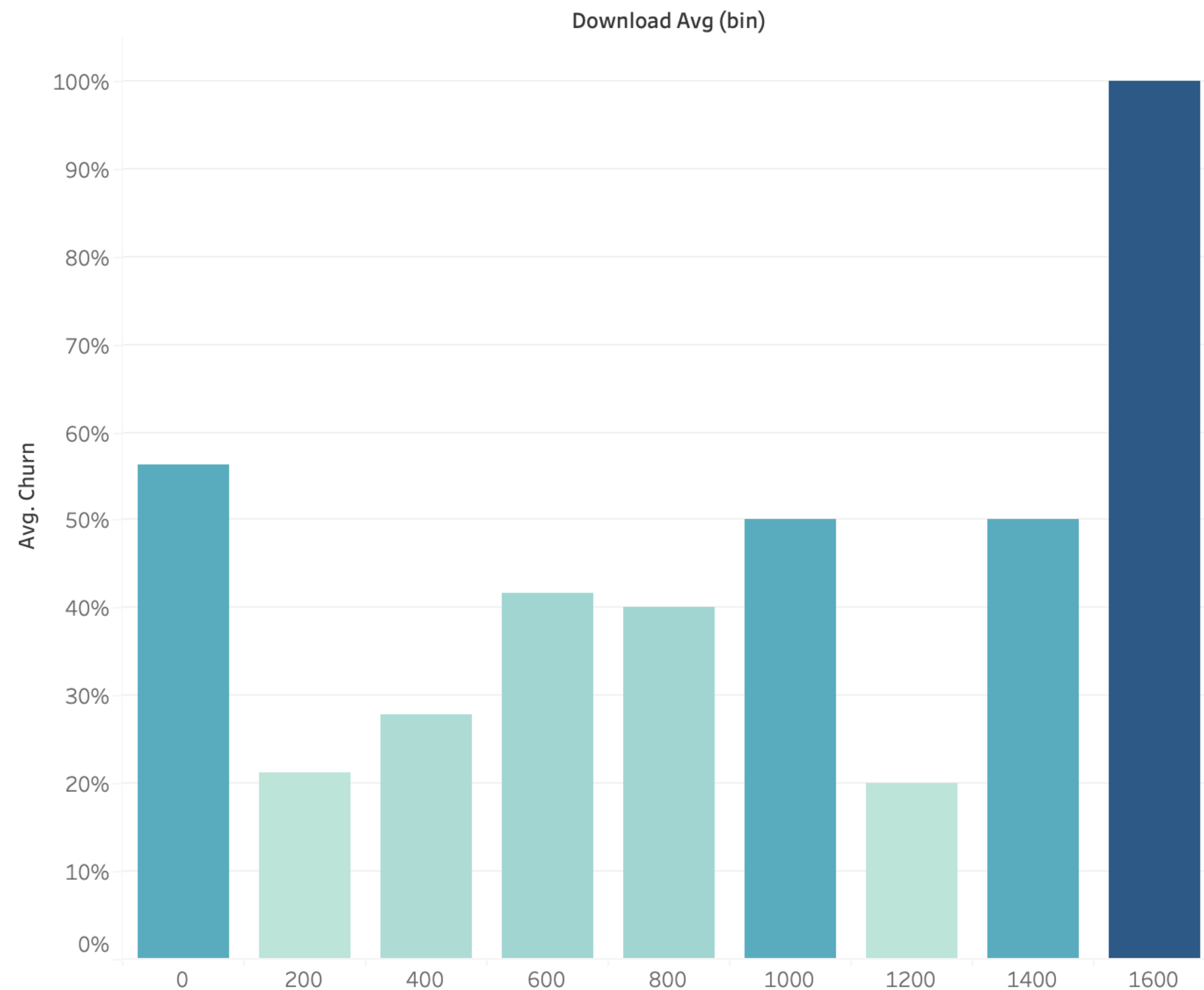
- ❖ Even with 0–3 service failures, churn is already very high (>90%).
- ❖ As failures become more frequent (4+), churn is almost 100%.

Business significance:

Customers are highly sensitive to service disruptions — even a small issue is enough to cause them to churn.

Churn Increases with Higher Download Usage

Churn Rate by Download Usage



Key Insight:

- ❖ Churn tends to increase with higher download usage.
- ❖ Customers with very high usage (highest bin) even approach 100% churn.
- ❖ Low-medium usage is relatively stable, but begins to rise as usage becomes heavier.

Business implications:

- ❖ Heavy users are the most frustrated → likely to experience frequent limits, throttling, or decreased network quality.
- ❖ Churn is not due to price, but because the user experience worsens with high usage.

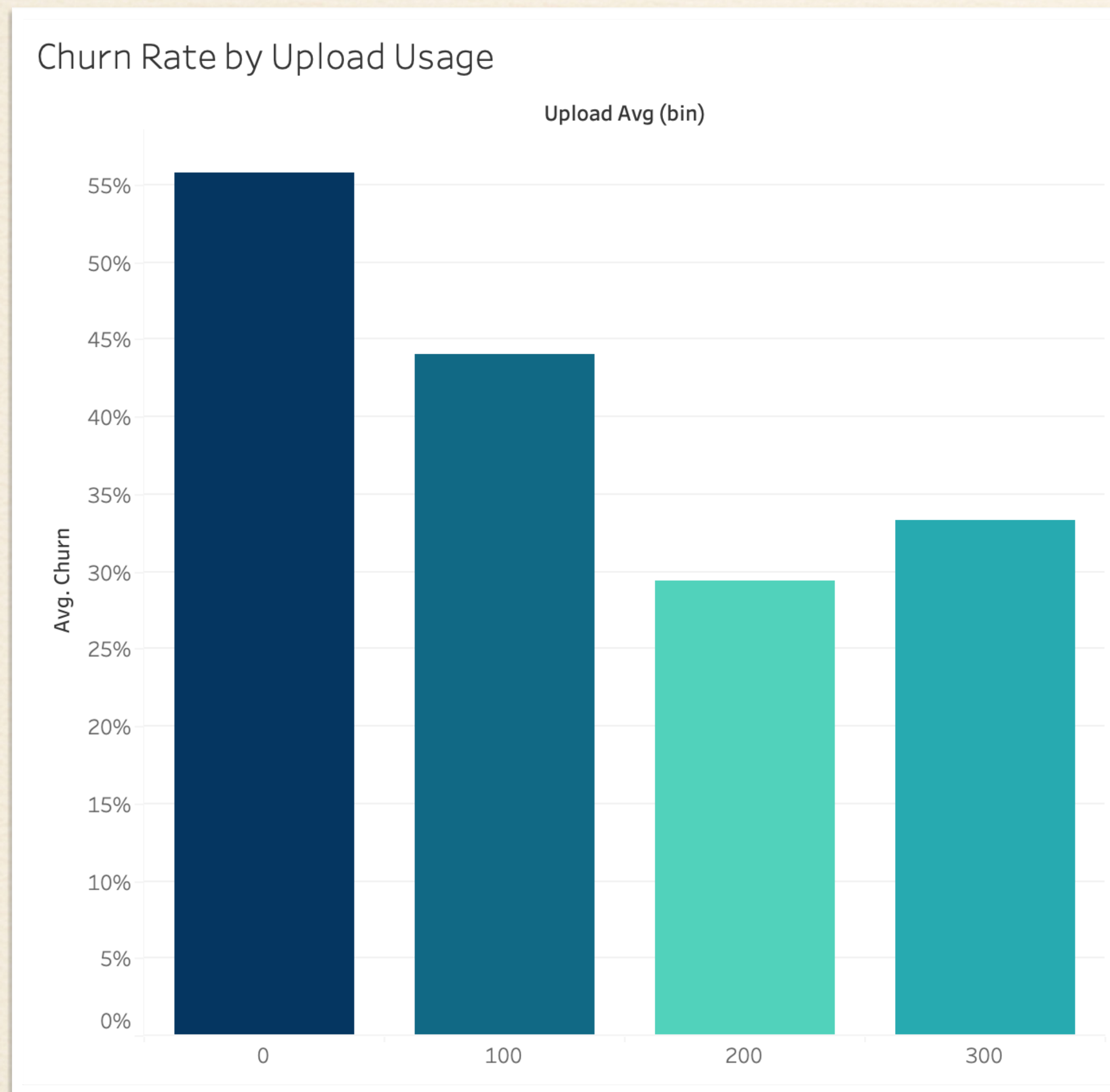
Churn Rate by Upload Usage

Key Insight:

- ❖ Churn is highest when upload usage is very low
- ❖ As upload usage increases, churn actually decreases
- ❖ It increases slightly again in the highest bin, but remains lower than low usage

Business significance:

- ❖ Customers who rarely upload are likely to have low user engagement → are more likely to churn
- ❖ Active users (who upload more frequently) tend to be more loyal



Key Churn Drivers

Biggest drivers of churn:

 Exceeding usage limits → sharp increase in churn

 Frequent service failures → nearly 100% churn

 Low usage (download & upload) → high churn

 Short contracts → highest churn

 Not taking a TV/Movie bundle → higher churn

Churn is primarily driven by usage friction, service reliability issues, and short-term customers—not pricing alone.

Key Findings & Recommendations

Key Findings	Business Recommendations
Objective: Reduce churn from 55.4% to <50% in 6 months.	Implement bundling upsell + over-limit policy redesign + early renewal campaigns.
Insight 1: Download over-limit is strongest churn driver (99.8% churn at 7x).	Reduce over-limit friction: early notifications, flexible add-on packages, transparent pricing.
Insight 2: TV/Movie bundling reduces churn dramatically (47.7% vs 89.6% without TV).	Prioritize bundling upsell to high-risk, heavy-usage segments; offer discounts for adoption.
Insight 3: Remaining contract strongly protects against churn (-2.66 coefficient).	Launch proactive early renewal campaigns 3 months before contract expiry; offer loyalty bonuses.
Insight 4: Service failures slightly elevate churn (0.29 vs 0.25).	Improve fault resolution SLA; provide automatic post-incident compensation/credits.
Problem Summary: Churn driven by experience (over-limit, failures, bundling) NOT price.	Recommendation: Focus on experience optimization over discounting; target top 20–30% risk segment.

Conclusion & Business Impact

Conclusion

- ❖ Churn is not driven by price, but by:
- ❖ Over-limit usage friction
- ❖ Lack of bundling value (TV/Movie)
- ❖ Contract renewal behavior
- ❖ Service experience

The biggest factor driving churn:

Download over-limit exceedance

The strongest factor inhibiting churn:

- ❖ Bundling subscription + remaining contract
- ❖  Business Impact (If Actions Implemented)
- ❖  Churn has the potential to decrease by 5–10% in 6 months
- ❖  Retention increases in high-usage segments

Stable revenue through:

- ❖ Upsell bundling
- ❖ Add-on packages
- ❖ Early renewal programs

Next Steps & Action Plan

Phase	Action	Owner	Timeline	Expected Impact
Short-term	Implement over-limit alerts & usage notifications	Product Team	Month 1	Reduce sudden churn spikes
Short-term	Launch bundling promo (TV + Movie + Internet)	Marketing Team	Month 1	Increase retention in high-usage users
Mid-term	Redesign flexible over-limit add-on packages	Product & Finance	Month 2–3	Lower friction & increase upsell
Mid-term	Early contract renewal campaign	CRM Team	Month 2–3	Improve retention before expiry
Long-term	Improve service failure resolution SLA	Operations Team	Month 4–6	Better customer experience

Next Steps & Action Plan

KPI (Optional small box under table)

- ❖ Churn rate < 50%
- ❖ Over-limit incidents decrease
- ❖ Bundling adoption increase
- ❖ Renewal rate increase

Final Conclusion / Key Takeaways

✓ Key Takeaways

- ❖ Customer churn is primarily driven by usage friction and service experience, not pricing.
- ❖ Over-limit usage is the strongest churn trigger across customers.
- ❖ Service failures significantly accelerate churn risk.
- ❖ Bundling services (TV & Movie packages) effectively reduces churn.
- ❖ Customers with short remaining contracts are the most vulnerable segment.

🎯 One-liner :

Improving customer experience and reducing usage friction are the most effective ways to lower churn.

Tools & Links (Deliverables)

Google Slide (M0-M4 deck) Link Source

Google Colab Notebook (Data Cleaning, EDA, Modeling) Link Source

Dataset source / file location (CSV) Link Source

Shared Folder (submission folder) Link

Tableau Link

Thank you...



Muhammad Nur Zhifar
Finish at February 01 2026
RevoU FSDA Batch OCT25
DEEPP